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Continuity

Limit of the Function $\frac{\sin \theta}{\theta}$ as $\theta \rightarrow 0$ Ling-Chan
YZU

- What is the area of the sector with central angle θ , where $\pi/2 > \theta > 0$?

$$\frac{1}{2} \cdot \pi \times \frac{\theta}{2\pi} = \frac{\theta}{2}$$

- What is the area of the triangle OTA ?

$$\frac{1}{2} \tan \theta$$

- We can clearly see that

$$\text{area}(\triangle OTA) \geq \text{area}(\text{sector } OPA)$$

- Therefore, we have

$$\frac{1}{2} \tan \theta \geq \frac{\theta}{2} \quad \underline{\tan \theta \geq \theta}$$

- Recall that we have shown that $\theta > \sin \theta$ for $\pi/2 > \theta > 0$
- Combining the previous two inequality, we obtain

$$\frac{\sin \theta}{\cos \theta} \geq \theta \quad \sin \theta \geq \theta \cdot \cos \theta$$

$$\theta \geq \sin \theta \geq \theta \cdot \cos \theta \xrightarrow{\div \theta} 1 \geq \frac{\sin \theta}{\theta} \geq \cos \theta$$

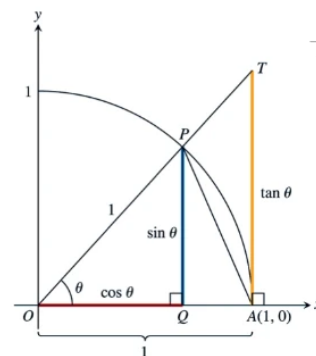


FIGURE 2.33 The ratio $TA/OA = \tan \theta$, and $OA = 1$, so $TA = \tan \theta$.

- We continue the derivation for $\pi/2 > \theta > 0$ that

$$\sin \theta \leq \theta \leq \tan \theta$$

$$\Leftrightarrow \frac{\sin \theta}{\sin \theta} \leq \frac{\theta}{\sin \theta} \leq \frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\sin \theta} \quad \text{if } \sin \theta > 0$$

$$\Leftrightarrow 1 \leq \frac{\theta}{\sin \theta} \leq \frac{1}{\cos \theta}$$

$$\Leftrightarrow 1 \geq \frac{\sin \theta}{\theta} \geq \cos \theta$$

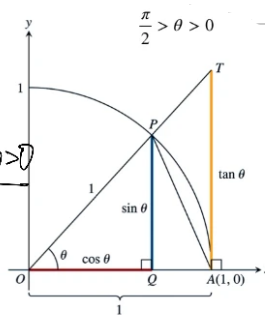


FIGURE 2.33 The ratio $TA/OA = \tan \theta$, and $OA = 1$, so $TA = \tan \theta$.

- Using the sandwich theorem, we have

$$\lim_{\theta \rightarrow 0^+} 1 \geq \lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} \geq \lim_{\theta \rightarrow 0^+} \cos \theta$$

$$\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1 \quad \lim_{\theta \rightarrow 0^+} \cos \theta = 1$$

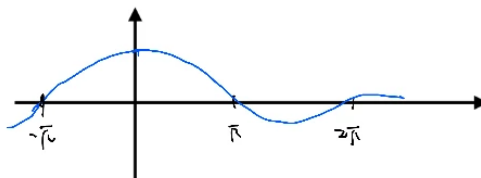
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The Sinc Function

- We can extend the domain to include $\theta = 0$
- We define a new function, called the sinc function as

$$\text{sinc}(\theta) \triangleq \begin{cases} \frac{\sin \theta}{\theta} & \text{if } \theta \neq 0 \\ 1 & \text{else if } \theta = 0 \end{cases}$$

- Is $\text{sinc}(\theta)$ continuous at 0? yes $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = \text{sinc}(0)$



Properties of Continuous Functions

- Given two functions f and g which are both continuous at c . Then normal algebraic combinations as below are also continuous at c .

- Sums: $f(x) + g(x)$

- Differences: $f(x) - g(x)$

- Constant multiples: $f(x) \cdot a$

- Products: $f(x) \cdot g(x)$

- Quotients: $\frac{f(x)}{g(x)}$

- Powers: $f^n(x)$

- Roots: $f^{\frac{1}{n}}(x)$